

DAY 4 HANDS-ON SUBNETTING PRACTICAL

Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

ANS:

To divide the network address 192.168.10.0/24 into 4 equal subnets, we need to borrow 2 host bits to use as subnet bits ($2^2 = 4$).

- **Original Subnet Mask:** /24 or 255.255.255.0
- **New Subnet Mask:** /26 or 255.255.255.192
- **Total Host IPs per Subnet:** $2^{(32-26)} = 64$ IPs
- **Usable Host IPs per Subnet:** $64 - 2 = 62$ usable hosts (excluding the network and broadcast addresses)

Here are the details for each of the 4 subnets:

Subnet 1

- **Subnet Mask:** 255.255.255.192 (/26)
- **Network Address:** 192.168.10.0
- **Usable IP Range:** 192.168.10.1 to 192.168.10.62
- **Broadcast Address:** 192.168.10.63

Subnet 2

- **Subnet Mask:** 255.255.255.192 (/26)
- **Network Address:** 192.168.10.64
- **Usable IP Range:** 192.168.10.65 to 192.168.10.126
- **Broadcast Address:** 192.168.10.127

Subnet 3

- **Subnet Mask:** 255.255.255.192 (/26)
- **Network Address:** 192.168.10.128
- **Usable IP Range:** 192.168.10.129 to 192.168.10.190
- **Broadcast Address:** 192.168.10.191

Subnet 4

- **Subnet Mask:** 255.255.255.192 (/26)
- **Network Address:** 192.168.10.192
- **Usable IP Range:** 192.168.10.193 to 192.168.10.254
- **Broadcast Address:** 192.168.10.255

For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.

ANS:

Here is the IP assignment using **Subnet 1** (192.168.10.0/26, usable range: 192.168.10.1 to 192.168.10.62):

Device	Assigned IP Address	Subnet Mask
Router (Default Gateway)	192.168.10.1	255.255.255.192 (/26)
Laptop 1	192.168.10.2	255.255.255.192 (/26)
Laptop 2	192.168.10.3	255.255.255.192 (/26)
Printer	192.168.10.4	255.255.255.192 (/26)
Mobile	192.168.10.5	255.255.255.192 (/26)

Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

ANS:

[Internet / WAN]

|
[Main Router / Firewall] (Gateway: 192.168.1.1)

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| (Port 1 / VLAN 10) | (Port 2 / VLAN 20) | (Port 3 / VLAN 30)
v v v
+-----+ +-----+ +-----+

Subnet: LAN	Subnet: Guest	Subnet: IoT
IP Range:	IP Range:	IP Range:
192.168.1.10 - .50	192.168.2.10 - .50	192.168.3.10 - .50
Connected Devices:	Connected Devices:	Connected Devices:
- Work Laptop	- Visitor Phones	- Smart TV
- Desktop PC	- Visitor Laptops	- Smart Lights
- Network Printer	- Security Camera	

Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

ANS:

Find Subnet Mask: To make 4 subnets, we borrowed 2 bits ($2^2 = 4$), turning our /24 prefix into a /26 mask (255.255.255.192).

Calculate Hosts: With 6 remaining host bits ($32 - 26 = 6$), each subnet gets $2^6 = 64$ total addresses, leaving **62 usable host IPs** (minus network and broadcast).

Determine Ranges (Block Size 64):

- Subnet 1: 192.168.10.0 – 192.168.10.63
- Subnet 2: 192.168.10.64 – 192.168.10.127
- Subnet 3: 192.168.10.128 – 192.168.10.191
- Subnet 4: 192.168.10.192 – 192.168.10.255

Assign Devices: Using Subnet 1, we safely allocated usable IPs: Router (.1), Laptop 1 (.2), Laptop 2 (.3), Printer (.4), and Mobile (.5).